SECP3723-01-SYSTEM DEVELOPMENT TECHNOLOGY

INDIVIDUAL PROJECT

SYSTEM: COURSE REGISTRATION SYSTEM

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System Introduction

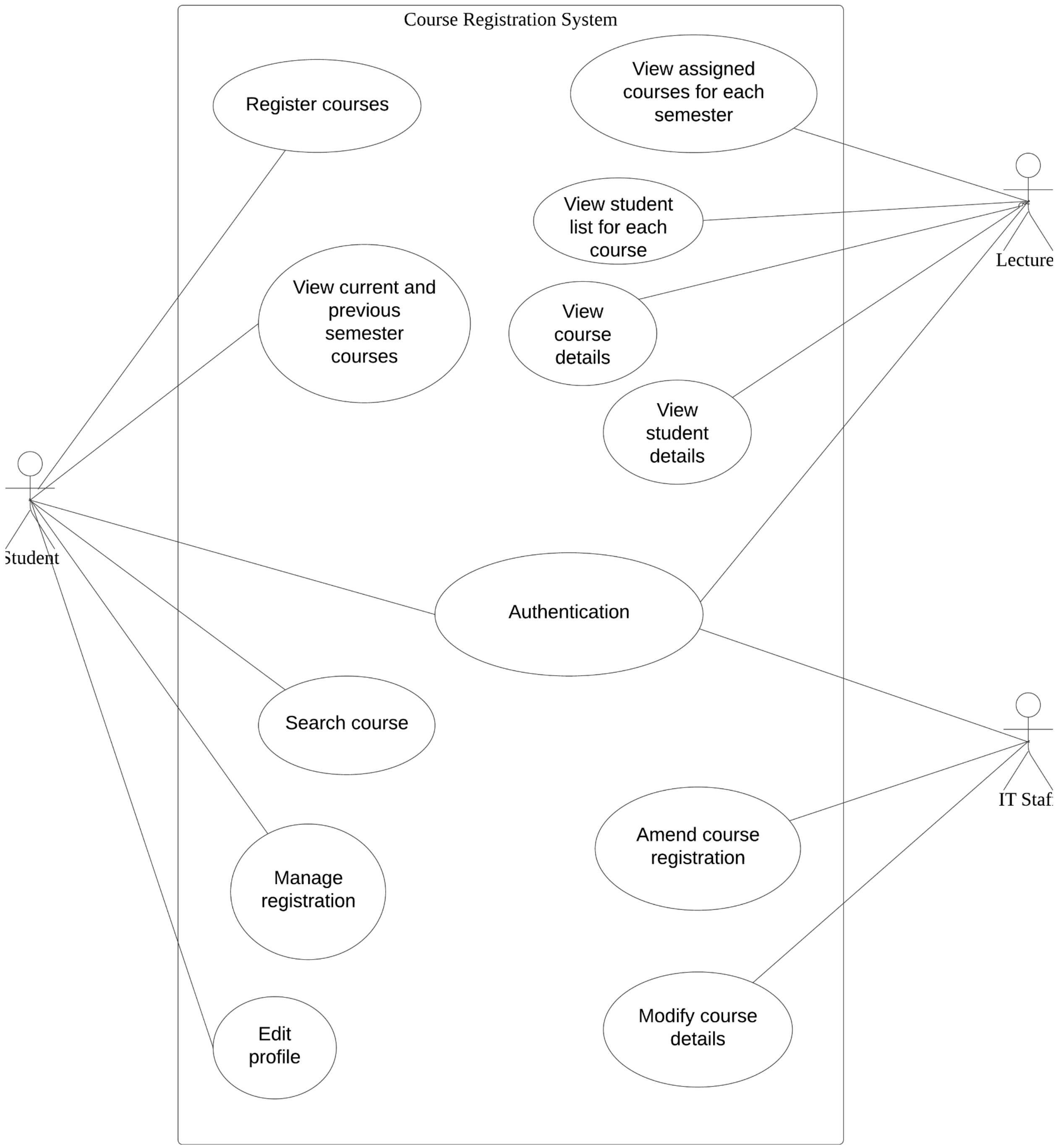
Course Registration is a web-based application designed to manage the process of course registration for students, lecturers, and IT staff in an institution. The system aims to provide an efficient platform for managing course enrollments, viewing course details, and handling administrative tasks related to course management.

The primary purpose of the system is to allow students to register for courses, view their current semester courses, search for available courses, and modify or cancel their registrations. When registering a course, it will update the current student in the database for the table courses. If the current student in the table course is less than the max student, it will auto approve the registration and the current student increment. Otherwise, the status is pending and the current student will not increment. When modifying the registration, it will also modify the current student for every course based on the status of the course. Lecturers can use the system to view the courses they are assigned to, access the list of students enrolled in their courses, and view detailed course information. IT staff (admin) have the ability to add, modify, or delete course details and manage student registrations.

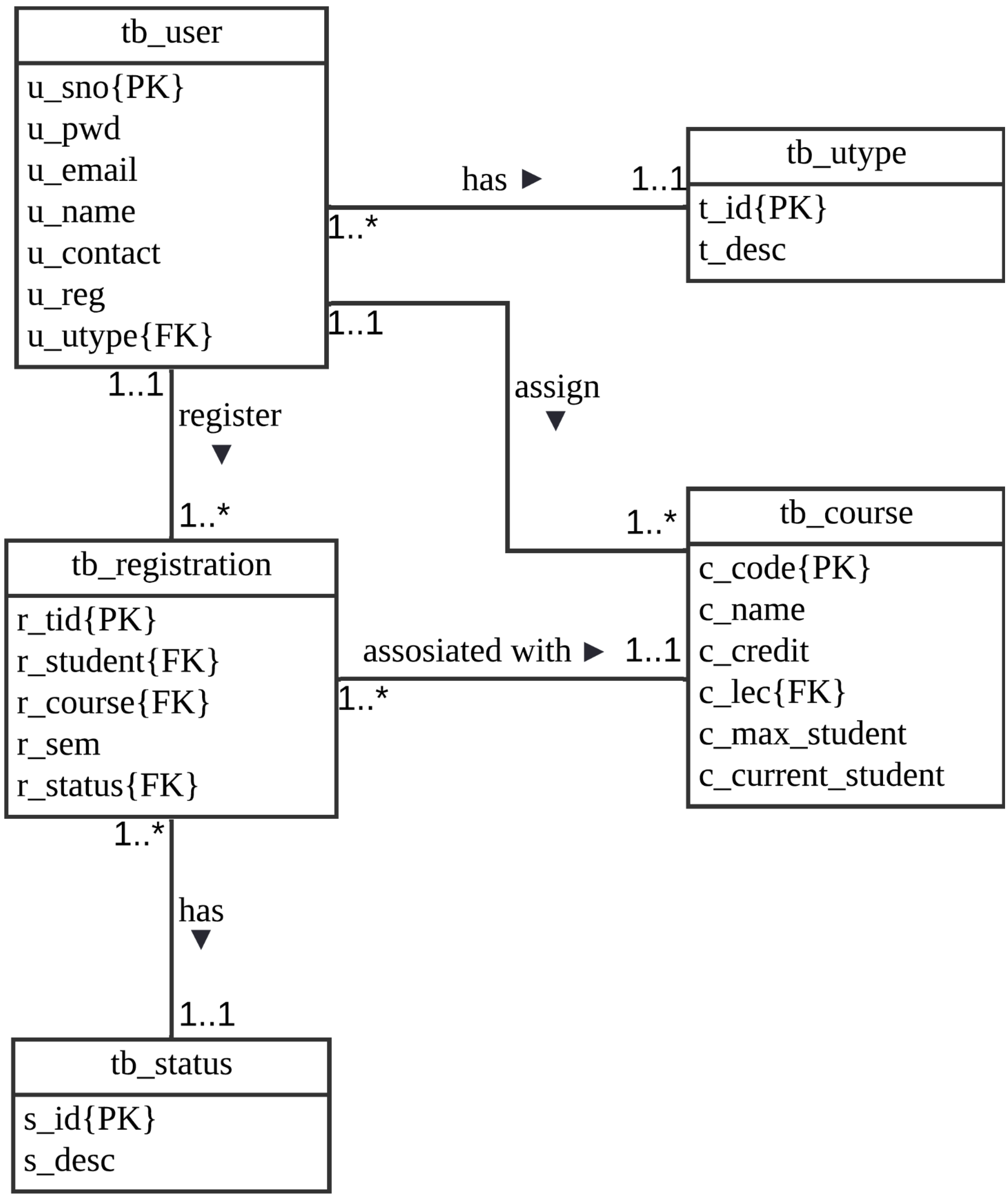
The Course Registration System is crucial for educational institutions because it automates the manual process of course registration and saves time for both students and staff. By providing a centralized online-based system for course management, the system enhances the overall performance of academic operations.

System Design

Use Case:



Logical ERD:



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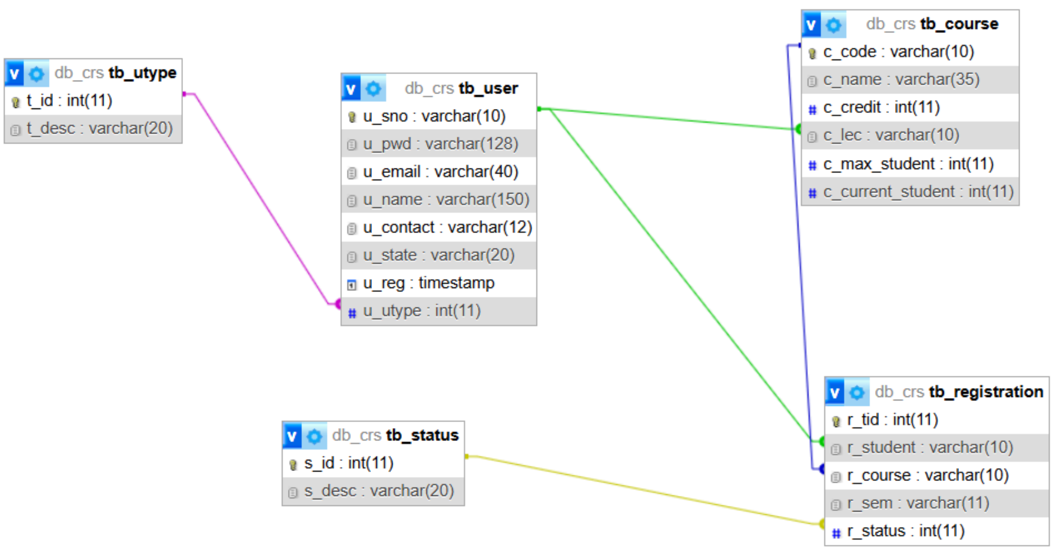
Development Software, Language, Technology and Tools

For the development software, XAMPP and VS Code have been used. XAMPP is used for local hosting and database setup which includes MySQL and phpMyadmin that are used for visualizing the database for easy to understand. VS Code is used for code development. Programming languages that have been used are PHP, Javascrip, HTML, CSS and SQL. Technologies like bootstrap 5 have also been used to make the user interface look more modern and clean with ease. Copilot also helped a lot on this project as it can help fix systaxs.

Database Design

Database design from phpMyAdmin



Development Step

Planning:

Requirement of development for this project has already been determined by the lecturer for this assessment . Sketches were created to visualize the user interface for all the users in the systems.

Design:

The database was designed using an ERD, and use case diagrams were created to visualize the requirement of this project. The system architecture was planned, and the UI/UX was designed to ensure a user-friendly experience.

Implementation:

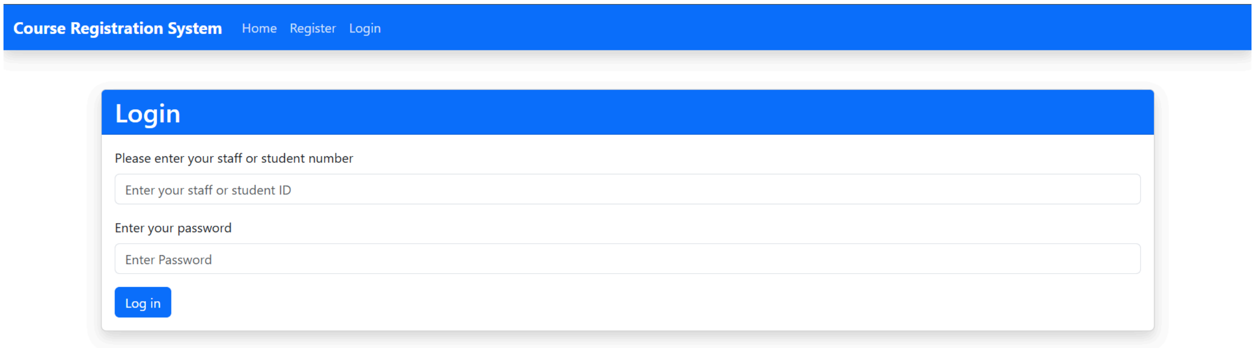
The back-end was developed using PHP, and the front-end was built with HTML, CSS, and JavaScript. The database was integrated using MySQL, and additional features like password hashing and SQL injection prevention were implemented.

Testing:

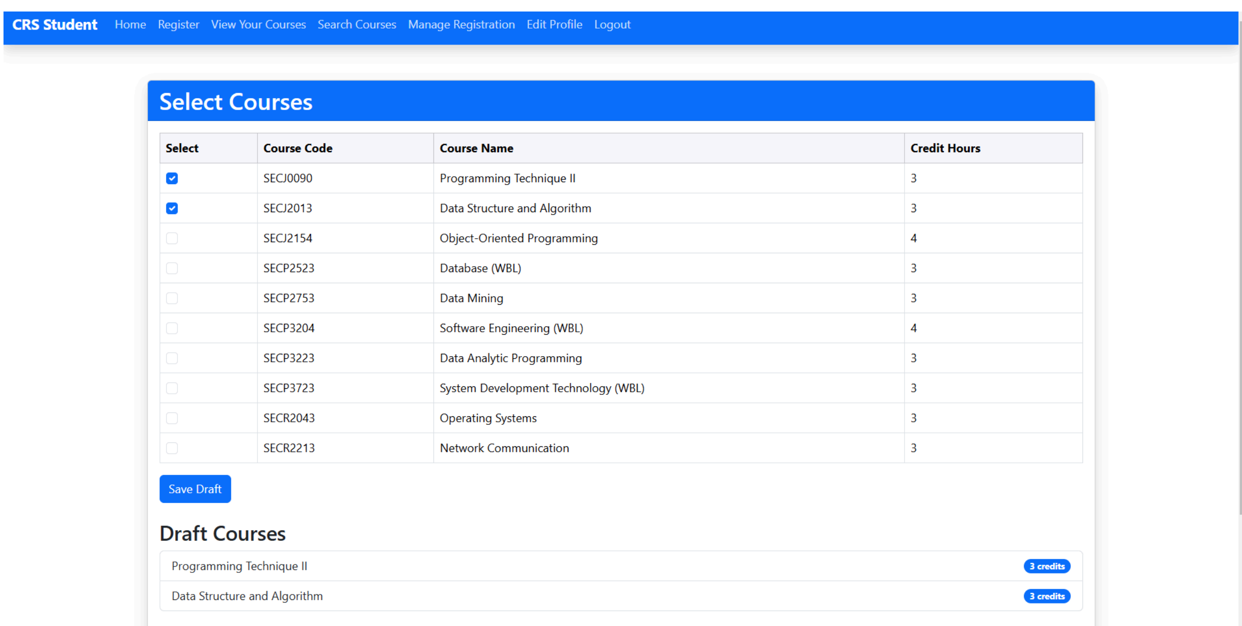
Unit testing and integration testing were conducted to ensure all components worked correctly. Project tested with dummy data for each user to interact with all the functions.

System Interface for All Users

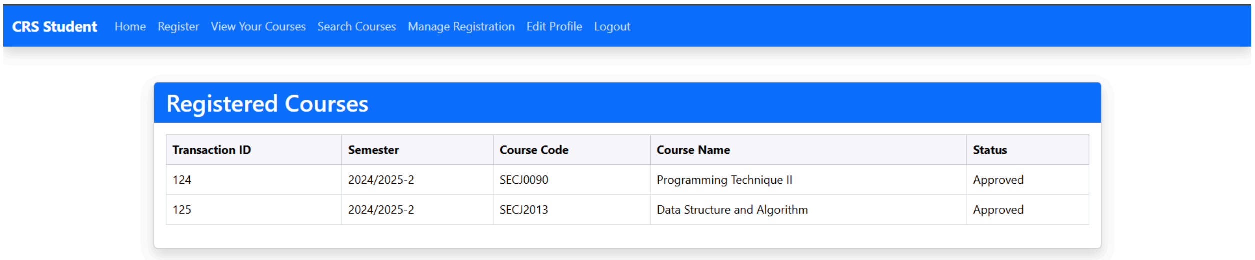
User interface for login page



User interface for student register course

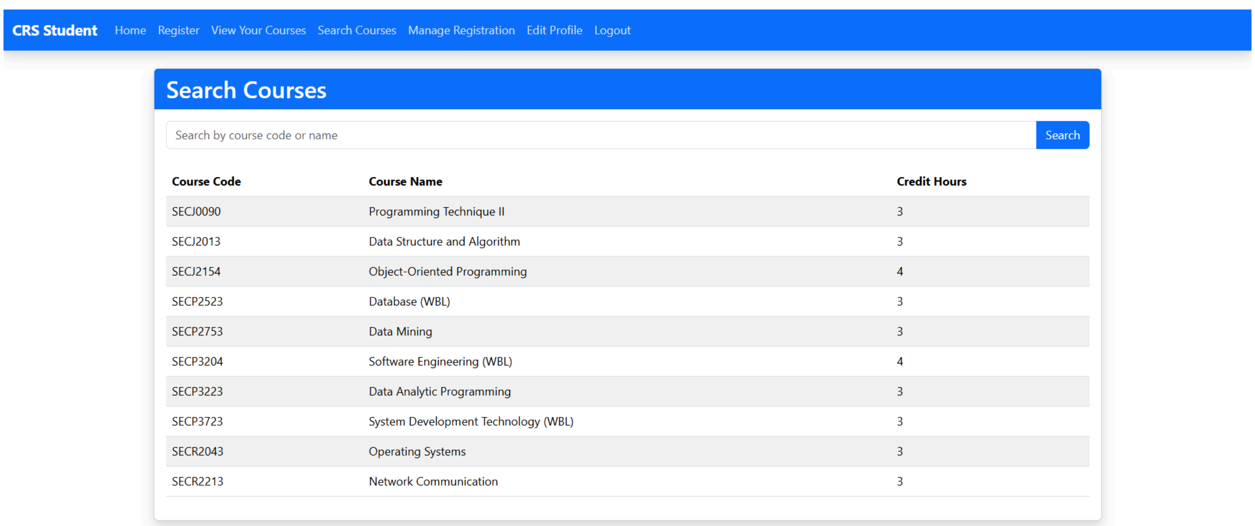


User interface for student view registered course



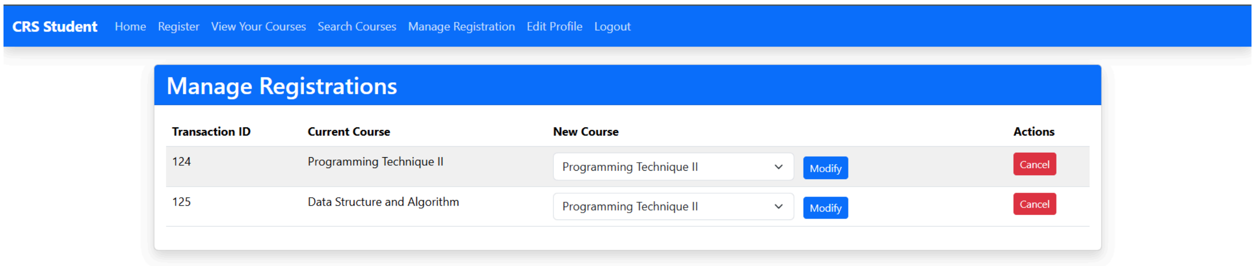
Transaction ID	Semester	Course Code	Course Name	Status
124	2024/2025-2	SECJ0090	Programming Technique II	Approved
125	2024/2025-2	SECJ2013	Data Structure and Algorithm	Approved

User interface for student search courses

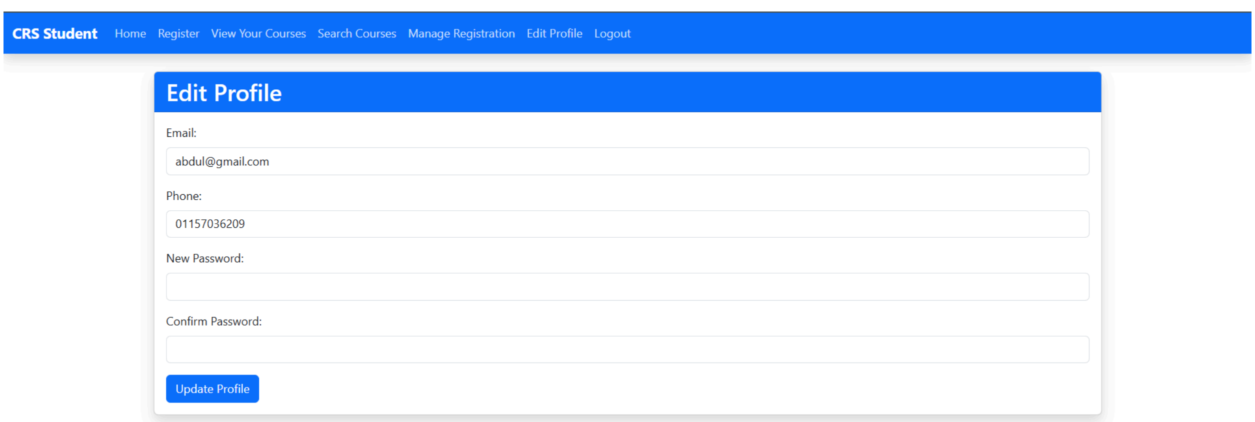


Course Code	Course Name	Credit Hours
SECJ0090	Programming Technique II	3
SECJ2013	Data Structure and Algorithm	3
SECJ2154	Object-Oriented Programming	4
SECP2523	Database (WBL)	3
SECP2753	Data Mining	3
SECP3204	Software Engineering (WBL)	4
SECP3223	Data Analytic Programming	3
SECP3723	System Development Technology (WBL)	3
SECR2043	Operating Systems	3
SECR2213	Network Communication	3

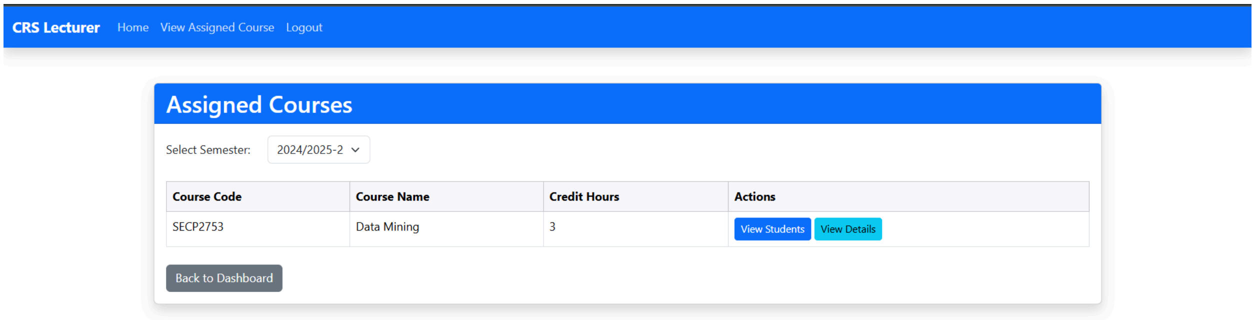
User interface for student manage registration



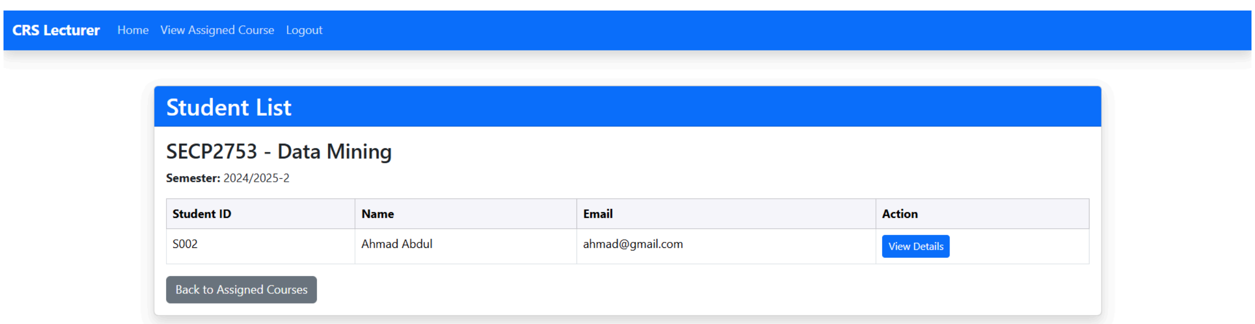
User interface for student edit profile



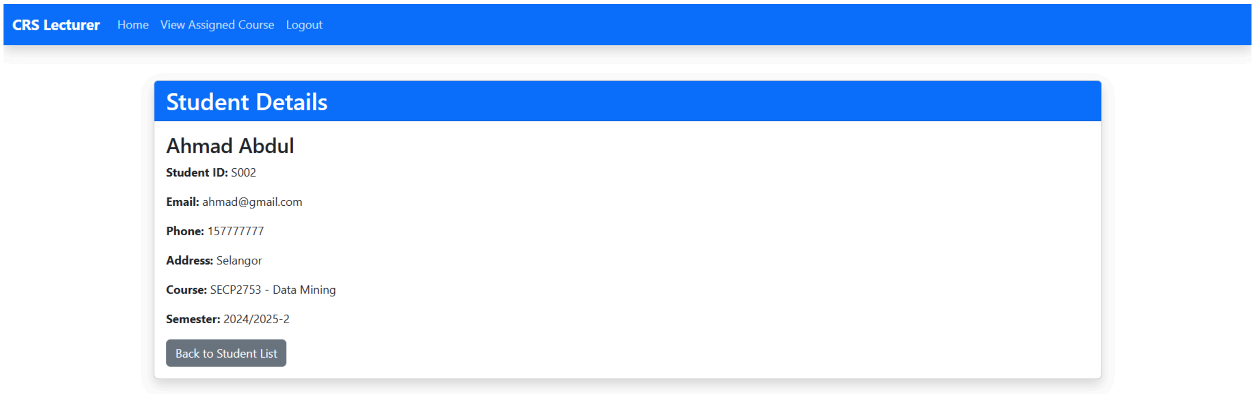
User interface for lecturer view assigned courses for each semester



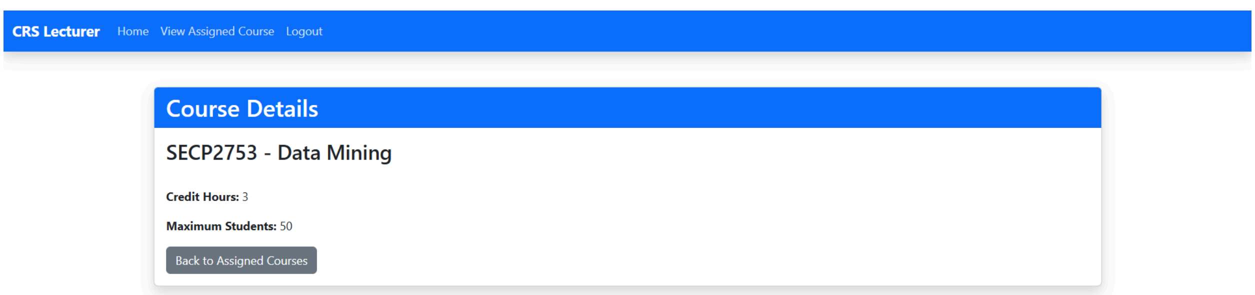
User interface for lecturer view student list for each course



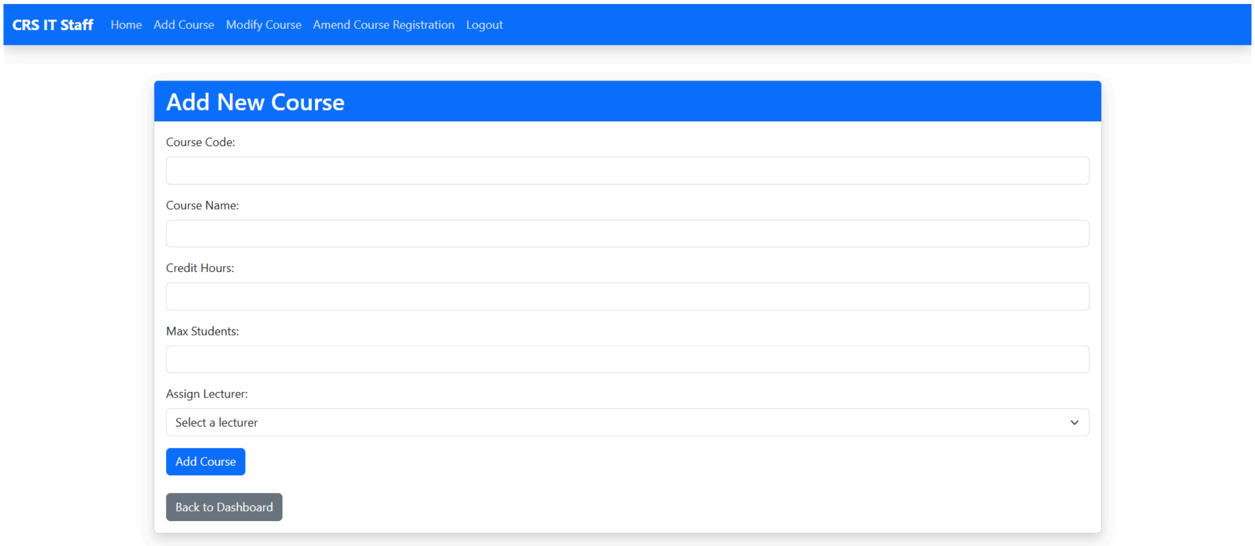
User interface for lecturer view student details



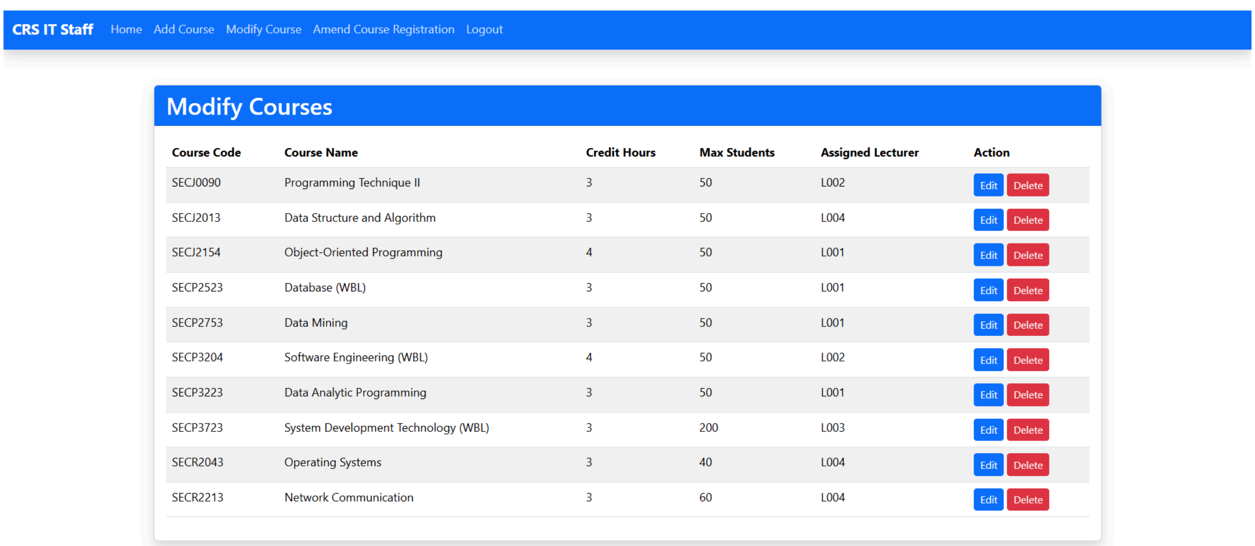
User interface for lecturer view course details



User interface for IT staff add new course

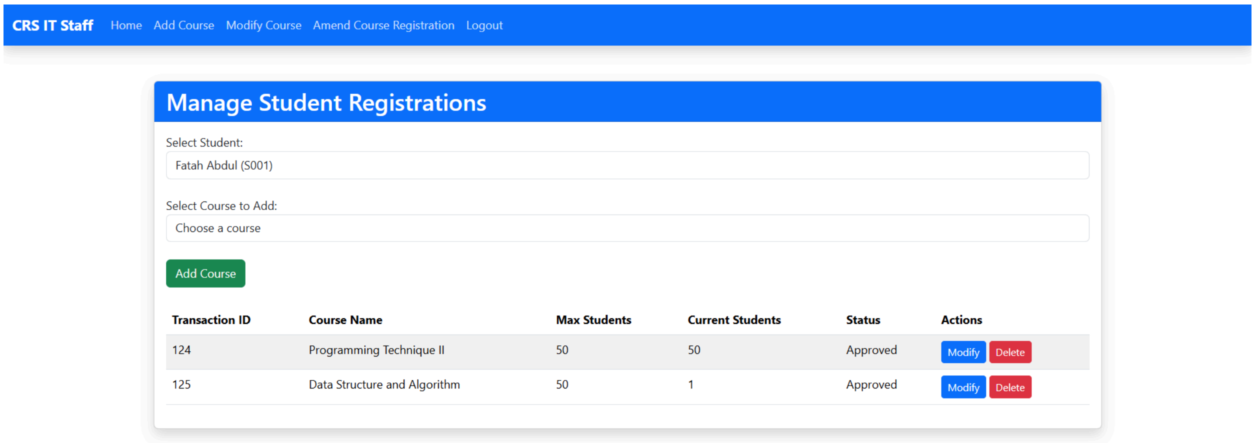


User interface for IT staff modify course

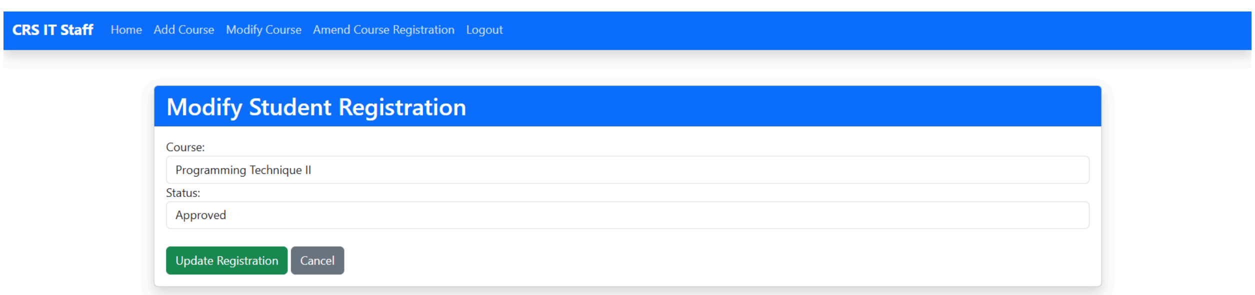


Course Code	Course Name	Credit Hours	Max Students	Assigned Lecturer	Action
SECJ0090	Programming Technique II	3	50	L002	Edit Delete
SECJ2013	Data Structure and Algorithm	3	50	L004	Edit Delete
SECJ2154	Object-Oriented Programming	4	50	L001	Edit Delete
SECP2523	Database (WBL)	3	50	L001	Edit Delete
SECP2753	Data Mining	3	50	L001	Edit Delete
SECP3204	Software Engineering (WBL)	4	50	L002	Edit Delete
SECP3223	Data Analytic Programming	3	50	L001	Edit Delete
SECP3723	System Development Technology (WBL)	3	200	L003	Edit Delete
SECR2043	Operating Systems	3	40	L004	Edit Delete
SECR2213	Network Communication	3	60	L004	Edit Delete

User interface for IT staff amend registration



User interface for IT staff amend registration (modify)



Localhost Setup

To set up the Course Registration System on a local machine, XAMPP, a popular local server environment that includes Apache, MySQL, and PHP has been used. The first step was to download and install XAMPP from the official website. After installation, the XAMPP Control Panel was launched, and the Apache and MySQL services were started. These services are essential for running the web server and database locally.

Next, the database for the system was set up using phpMyAdmin, which is included in XAMPP. PhpMyAdmin can be accessed by navigating to http://localhost/phpmyadmin in a web browser. A new database named db_crs was created, and designed inside it.

After setting up the database, the system files were placed in the htdocs folder within the XAMPP installation directory. For better organization, a new folder named crs was created inside htdocs, and all system files were created into this folder. This allowed the system to be accessed via the URL http://localhost/crs in a web browser.

After that, dbconnect.php has been created to connect the system files to the databases. In summary, the local setup process involved installing XAMPP, placing the system files in the htdocs folder, and configuring the files to connect to the database. This setup allowed for efficient development and testing of the Course Registration System before deployment to a live hosting platform.

User credentials

To demonstrate and test the Course Registration System, the following test accounts have been created for each user role. These credentials are valid and can be used to log in to the system:

Student Account:
Username: S001
Password: admin123

This account allows students to register for courses, view their current and previous semester courses, search for available courses, and modify or cancel their registrations.

Lecturer Account:
Username: L001
Password: admin123

This account enables lecturers to view the courses they are assigned to, access the list of students enrolled in their courses, and view detailed course information.

IT Staff (Admin) Account:
Username: T001
Password: admin123

This account provides IT staff with administrative privileges to add, modify, or delete course details, manage student registrations, and oversee the overall functionality of the system.

These test accounts have been carefully configured to ensure they work seamlessly with the system. Each account has been assigned the appropriate permissions and roles, allowing users to perform their respective functions without any issues.